

BLAST®

Basic Local Alignment Search Tool

[NCBI/ BLAST Home](#)

BLAST finds regions of similarity between biological sequences.

[more...](#)

New DELTA-BLAST, a more sensitive protein-protein search

BLAST Assembled Genomes

Find Genomic BLAST pages:

Enter organism common name, scientific name, or tax id.

- [Human](#)
- [Mouse](#)
- [Rat](#)
- [Cow](#)
- [Pig](#)
- [Dog](#)

- [Rabbit](#)
- [Chimp](#)
- [Guinea pig](#)
- [Fruit fly](#)
- [Honey bee](#)
- [Chicken](#)

- [Zebrafish](#)
- [Clawed frog](#)
- [Arabidopsis](#)
- [Rice](#)
- [Yeast](#)
- [Microbes](#)

Basic BLAST

Choose a BLAST program to run.

Program	Program Description
nucleotide blast	Search a nucleotide database using a nucleotide query <i>Algorithms:</i> blastn, megablast, discontinuous megablast
protein blast	Search protein database using a protein query <i>Algorithms:</i> blastp, psi-blast, phi-blast, delta-blast
blastx	Search protein database using a translated nucleotide query
tblastn	Search translated nucleotide database using a protein query
tblastx	Search translated nucleotide database using a translated nucleotide query

Specialized BLAST

Choose a type of specialized search (or database name in parentheses.)

- Make specific primers with [Primer-BLAST](#)
- Cluster multiple sequences together with their database neighbors using [MOLE-BLAST](#)
- Find [conserved domains](#) in your sequence (cds)
- Find sequences with similar [conserved domain architecture](#) (cdart)
- Search sequences that have [gene expression profiles](#) (GEO)
- Search [immunoglobulins and T cell receptor sequences](#) (IgBLAST)
- Screen sequence for [vector contamination](#) (vecscreen)

- [Align](#) two (or more) sequences using BLAST (bl2seq)
- Search [protein](#) or [nucleotide](#) targets in PubChem BioAssay
- Search [SRA by experiment](#)
- Constraint Based Protein [Multiple Alignment Tool](#)
- Needleman-Wunsch [Global Sequence Alignment Tool](#)
- Search [RefSeqGene](#)
- Search [trace archives](#)
- Search bacterial and fungal rRNA sequences with [Targeted Loci BLAST](#)

Your Recent Results New!

[All Recent results...](#)

News

[BLAST XML](#)

The BLAST+ 2.2.31 executables can produce a new version of the BLAST XML (XML2).

Fri, 10 Jul 2015 18:00:00 EST

[More BLAST news...](#)

Tip of the Day

[How to Search Custom Databases in Web-Blast Using Entrez Queries.](#)

A powerful feature of the BLAST Web interface is the ability to limit BLAST searches to a subset of any database using a standard Entrez query.

[More tips...](#)